



Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., III & ECE A

Course Code & Title: EC8392 & Digital Electronics

Name of the Faculty member (s): Mr.S.Vijayakumar

Innovative Practice Description

- **Unit / Topic:** III / Flip Flops – SR, D, JK and T
- **Course Outcome:** CO : The students will be able to Analyse and design synchronous sequential circuits
- **Topic Learning Outcome:** TLO : Explain the SR, JK, T, D, Master/Slave Flip-flops operation and its Triggering
- **Activity Chosen:** Jigsaw
- **Justification:**
 - The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal; each person depends on all the others.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO2	PSO3
CO3	1	2	2	2	2	2	2	2	3

- **PO / PSO mapped:**

Innovative practice	PO8	PO9	PSO3
	2	2	3
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	The concept of memory and programmable logic devices and logic families is required for design, analyze VLSI systems. So it is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students feel that it was interesting. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ <https://www.readingrockets.org/strategies/jigsaw>
- ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>



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Innovative Practice Description

- **Unit / Topic:** V/ Memory Devices
- **Course Outcome:** CO : The students will be able to explain the concepts of memory devices, digital integrated circuits and their characteristics
- **Topic Learning Outcome:** TLO: Explain the different basic memory structure
- **Activity Chosen:** Flipped class room
- **Justification:** It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Students are asked to prepare more in depth than before. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO2	PSO3
CO3	1	2	2	2	2	2	2	2	3

• **PO / PSO mapped:**

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- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students feel that they have improved self-learning. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

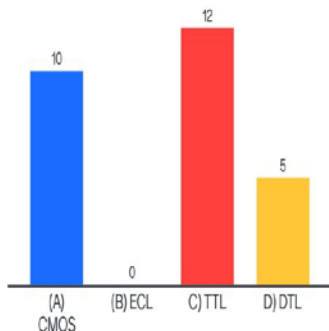
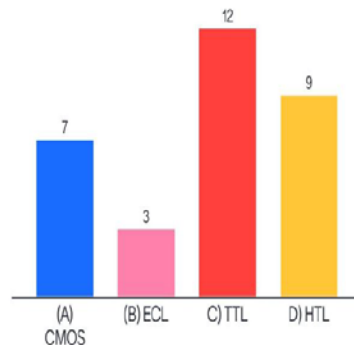
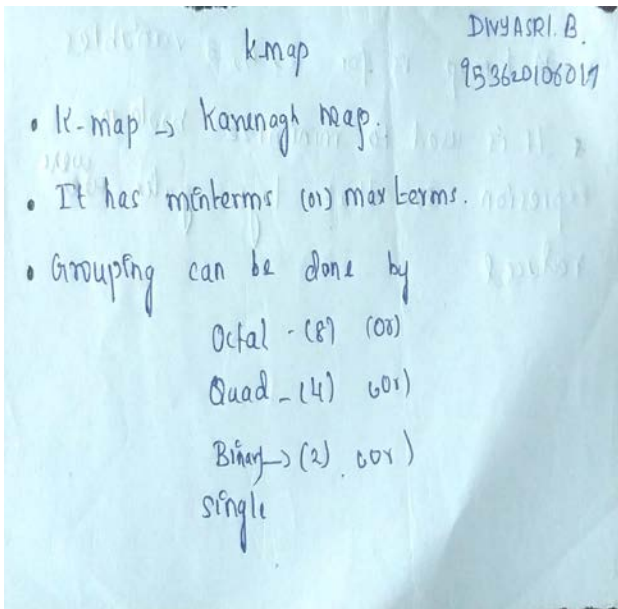
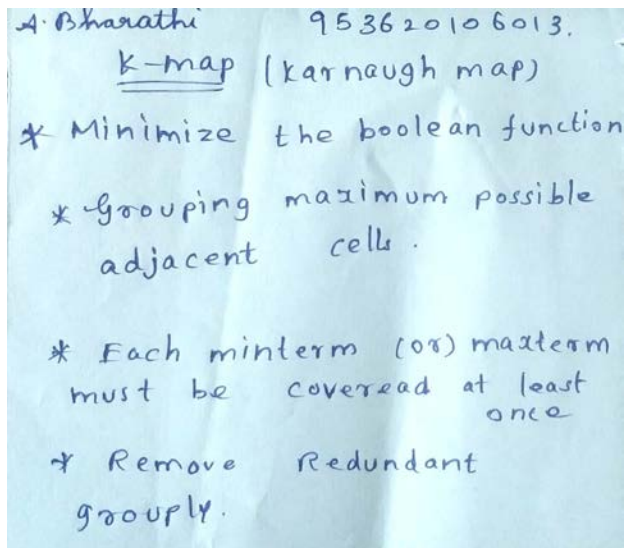
The main challenge faced is that few students not exposed to flipped class room. Students struggle with self-discipline and may turn up to class without having absorbed the lesson. To make all students to participate I started to praise others, taking turns for equal participation, and shared decision making.

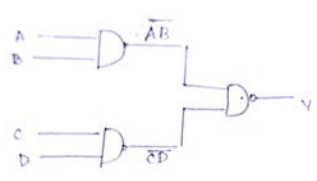
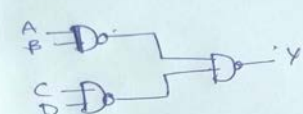
References:

- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.

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INNOVATIVE TEACHING METHOD PLAN

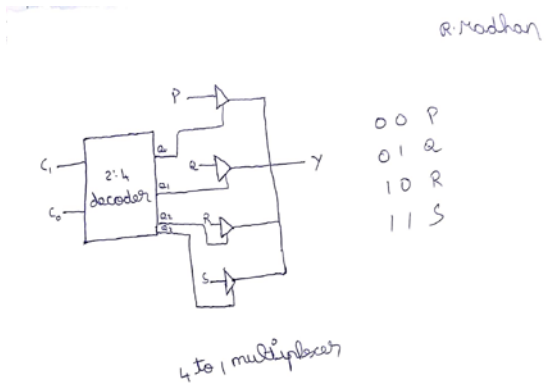
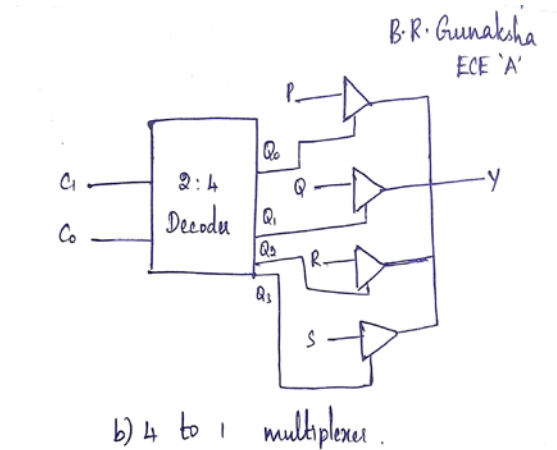
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UNIT I																							
1.	31.8.21	Logic gates and Universal gates	Class poll																				
Which of the following is the most popular family of SSI and MSI?  <table><thead><tr><th>Family</th><th>Count</th></tr></thead><tbody><tr><td>(A) CMOS</td><td>10</td></tr><tr><td>(B) ECL</td><td>0</td></tr><tr><td>(C) TTL</td><td>12</td></tr><tr><td>(D) DTL</td><td>5</td></tr></tbody></table>		Family	Count	(A) CMOS	10	(B) ECL	0	(C) TTL	12	(D) DTL	5	Which logic gate used in electric motors on-off, control circuit, high voltage switch?  <table><thead><tr><th>Gate</th><th>Count</th></tr></thead><tbody><tr><td>(A) CMOS</td><td>7</td></tr><tr><td>(B) ECL</td><td>3</td></tr><tr><td>(C) TTL</td><td>12</td></tr><tr><td>(D) HTL</td><td>9</td></tr></tbody></table>		Gate	Count	(A) CMOS	7	(B) ECL	3	(C) TTL	12	(D) HTL	9
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2	04.09.21	Karnaugh map Minimization, Don't care conditions	One minute paper																				
																							

3	07.09.21	Quine – Mc Cluskey method of minimization and Revision	Think aloud pair problem solving																																				
		Quine - McCluskey Method B.P. Gunakula R. Nethayarajam  $Y = ABCD$ <table border="1" data-bbox="285 759 572 916"> <tr> <td></td> <td>BC</td> <td>00</td> <td>01</td> <td>11</td> <td>10</td> </tr> <tr> <td>A</td> <td>0</td> <td>1</td> <td>0</td> <td>0</td> <td>1</td> </tr> <tr> <td>1</td> <td>1</td> <td>1</td> <td>1</td> <td>1</td> <td>1</td> </tr> </table> $Y = A + \bar{C}$		BC	00	01	11	10	A	0	1	0	0	1	1	1	1	1	1	1	Gr. Gokul Prinja M. Krimpa. 1.  <u>Ans:</u> ABC $\bar{A}\bar{B} \cdot \bar{C}\bar{D} \Rightarrow Y = ABCD$ <table border="1" data-bbox="780 602 1147 759"> <tr> <td></td> <td>BC</td> <td>00</td> <td>01</td> <td>11</td> <td>10</td> </tr> <tr> <td>A</td> <td>0</td> <td>1</td> <td>0</td> <td>0</td> <td>1</td> </tr> <tr> <td>1</td> <td>1</td> <td>1</td> <td>1</td> <td>1</td> <td>1</td> </tr> </table> <u>Ans:</u> $Y = A + \bar{C}$		BC	00	01	11	10	A	0	1	0	0	1	1	1	1	1	1	1
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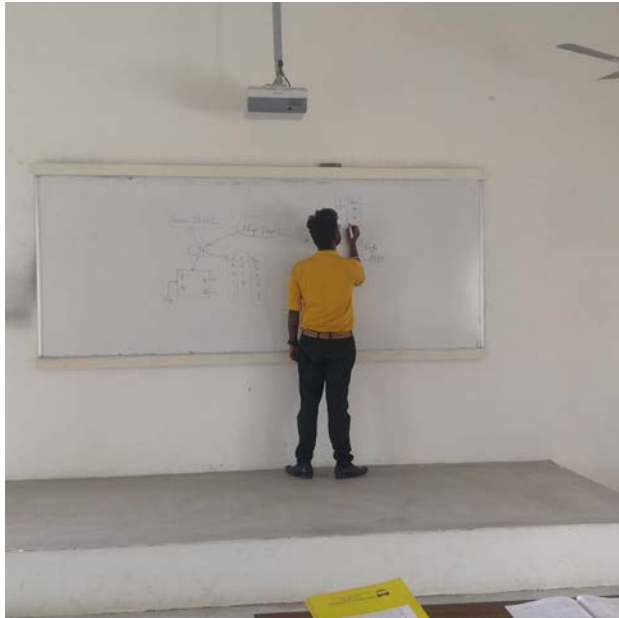
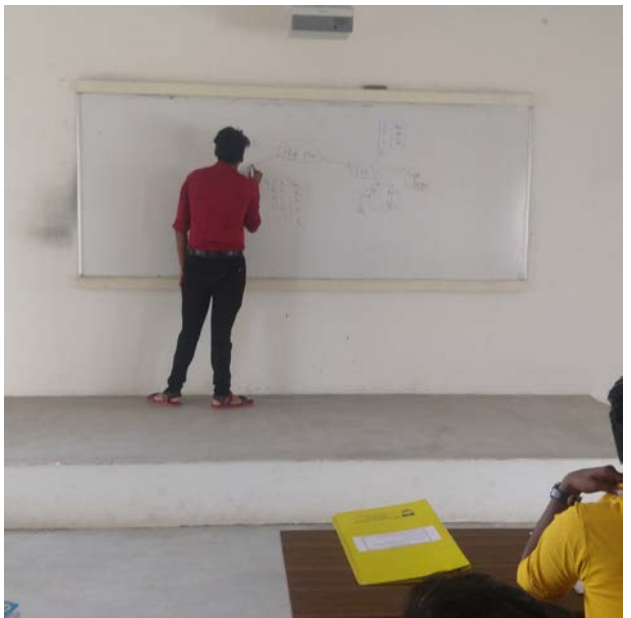
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UNIT II			
1	30.09.21	Multiplexer, Demultiplexer	Open book test
			
2	05.10.21	Decoder, Encoder, priority encoder	Note check
 R. Madhan 00 P 01 Q 10 R 11 S		 B.R. Gunaksha ECE 'A' b) 4 to 1 multiplexer.	

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UNIT III			
1	09.10.21	Flip Flops-JK, T	Mind Map
			
2	21.10.21	Design- Moore & Mealy models	Zero-Minute Speech
			

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UNIT IV			
1	27.10.21	Stable and Unstable states, Output specification	Exit slip
<u>Asynchronous Sequential circuit</u> E. Jackson * The change in input signals can affect memory element at any instant of time. * Because of the absence of clock, it can operate faster than synchronous circuit		<u>Asynchronous Sequential Circuit: ECE-A</u> * In asynchronous sequential circuit the change in input signal can affect the memory element at any instant of time. i.e. * Memory elements are either unlocked flip-flops or time delay elements	
2	02.11.21	Hazards, Essential Hazards	Think Pair Share
			

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UNIT V			
1	13.11.21	PROM, EPROM, EEPROM, EAPROM	Role play
			
2	21.11.21	Logic families and their characteristics	Visible quiz
		